

Westravelle Analytical Ltd, a Tarnovell company

CVL tablet lot 3: twelve-month ICH stability with assay validation

Final report — QA approved

Westravelle Analytical Ltd, a Tarnovell company

Report no. E-R-2021-025-01

Project: WA/2020/313 | LIMS: WST-20-009708

Client: Corvenlea Pharmaceuticals plc

Report date: 02 Dec 2021

Scientific lead: Dr A. Penrose

1 Approved summary

WA/2020/313 reports ICH chamber programme and Assay and related-substances validation for Corvenlea Pharmaceuticals plc: CVL tablet lot 3. Work began on 30 Nov 2020; the completed evidence set was closed on 30 Nov 2021. The tablet lot remained within the agreed assay and impurity limits, and Corvenlea Pharmaceuticals plc used the report in its formulation transfer package.

Dr A. Penrose authorises this final report on 02 Dec 2021. The key table retains the approved endpoint labels and units. Conclusions apply to the material, method and conditions tested. This issue supersedes working interpretations for the completed protocol; it does not authorise an altered product configuration or analytical procedure.

Table 1. Approved stability results

Condition	Timepoint (months)	Assay (% LC)	Imp A RRT 0.86 (%)	Imp B RRT 1.12 (%)	Total impurities (%)	Water KF (% w/w)	Result
25 °C/60% RH	0	100.6	0.04	<0.05	0.19	1.8	Complies
25 °C/60% RH	1	100.5	0.05	<0.05	0.20	1.6	Complies
25 °C/60% RH	3	100.1	0.08	<0.05	0.18	0.9	Complies
25 °C/60% RH	6	99.9	0.10	0.06	0.22	2.0	Complies
25 °C/60% RH	9	99.5	0.12	0.07	0.35	1.3	Complies
25 °C/60% RH	12	99.8	0.16	0.08	0.33	2.0	Complies
30 °C/65% RH	0	100.6	0.04	<0.05	0.19	1.8	Complies
30 °C/65% RH	1	100.3	0.06	<0.05	0.21	1.4	Complies
30 °C/65% RH	3	100.6	0.08	0.05	0.23	2.5	Complies
30 °C/65% RH	6	100.3	0.13	0.10	0.32	2.2	Complies
30 °C/65% RH	9	100.3	0.19	0.10	0.36	0.9	Complies
30 °C/65% RH	12	99.6	0.21	0.11	0.45	1.9	Complies
40 °C/75% RH	0	100.6	0.04	<0.05	0.19	1.8	Complies
40 °C/75% RH	1	99.8	0.10	<0.05	0.20	2.5	Complies
40 °C/75% RH	3	99.6	0.19	0.12	0.39	1.5	Complies
40 °C/75% RH	6	98.4	0.34	0.21	0.69	0.9	Complies

Specification: assay 95.0-105.0% LC; Imp A (RRT 0.86, specified) $\leq 0.50\%$; Imp B (RRT 1.12, specified) $\leq 0.30\%$; any unspecified impurity $\leq 0.20\%$; total impurities $\leq 2.0\%$

2 Protocol identity and responsibility

Project WA/2020/313, accession WST-20-009708, is the reference for receipt, preparation, acquisition and reporting. The client name in this approved issue is Corvenlea Pharmaceuticals plc. Product identity remains CVL tablet lot 3; an informal label does not replace the accession. The protocol addresses ICH chamber programme and Assay and related-substances validation.

Alice Penrose holds scientific responsibility. Leon Wu maintains sample and storage linkage. Saira Mahmood checks the procedure and system suitability assessment. At the release gate, the final endpoint must trace to both the material selected for analysis and the eligible sequence that generated it.

Responsibilities at final release

Responsibility	Person	Release evidence
Scientific conclusion	Dr A. Penrose	Approved report
Sample traceability	Dr L. Wu	Batch card and handling trail
Analytical review	Saira Mahmood	Method and sequence sheets

3 Objective and decision boundary

The transfer team required a linked account of lot identity, storage history and the suitability of the assay and related-substances procedure. This report supplies that account for CVL tablet lot 3. A transfer decision must retain the tested formulation and container configuration. A new formulation does not inherit the conclusion merely because its active ingredient is unchanged.

The acceptance question is settled against the agreed endpoint definitions. Meeting an analytical criterion is not automatically a general product-quality statement. Material identity, exposure history and procedure remain part of the decision. No clinical interpretation or commercial forecast is included. The report gives the evidence and its boundary so that the client can retain both when using the conclusion.

4 Chronology and release gates

Work started on 30 Nov 2020. The interim evidence point was 31 May 2021; work concluded on 30 Nov 2021. Final reporting occurred on 02 Dec 2021. These dates describe the protocol chronology and do not replace actual acquisition times on individual sequence sheets.

Interim interpretation was limited to the available evidence. Final release incorporated the completed evidence set and review of sample linkage and suitability. Earlier extracts preserve chronology but do not supersede this issue. A copied sequence date should remain its original date; resetting it to the report approval date would lose the relationship between preparation, acquisition and interpretation.

5 Receipt and accession control

Material was accepted using the client product designation and the assigned accession. Supplied labels, container description and intended analytical use were linked on the batch card. A local aliquot identifier remained subordinate to that accession. Identity was taken from the client documentation; an independent structural identity investigation was outside the protocol.

The file distinguishes material as received from preparations made for injection. A prepared solution is not a new stability lot. Labels were checked before opening material and again at preparation. This prevents an apparently eligible chromatogram from being reported under the wrong product description. The accession is retained on the preparation trail as well as the report.

6 Batch card and sample genealogy

The batch card links receipt, handling and preparation. Original product designations were retained when material was split. Each sequence sheet identifies the preparation, its source and its intended analytical purpose. Repeated injection remains linked to the existing preparation; a fresh preparation has its own entry.

Review followed the chain from receipt to reported endpoint, then traced the endpoint back to its source. A short chromatogram label alone cannot establish this chain. The report carries the project and accession so that an extracted page can be returned to the right file. Product identity is retained throughout the chain, including at the final calculation step.

7 Storage and handling before analysis

Material remained under its assigned handling conditions until removal or preparation. Removed material was distinguished from retained reserve. Time outside controlled storage and the preparation sequence belong to the sample history; they are not discarded because the eventual analytical result is satisfactory.

Containers were checked for label legibility, closure integrity and sufficient material. A compromised container would require a deviation note and impact decision before inclusion. Handling observations remain with the batch card, allowing later review to distinguish a change in the product from an event introduced during sampling. The report does not reconstruct an unrecorded temperature history.

8 References and reagent preparation

Reference preparations establish the response used for calculation. Preparation entries retain the reference designation, applicable potency correction and dilution trail. Reference potency was not inferred from the observed peak area. Blank, reference and sample preparations were identified separately so that their distinct purposes remained visible.

Reagent and mobile-phase labels were checked against the sequence sheet. A replacement preparation was linked to its actual point of use rather than silently inheriting its predecessor identity. The controlled preparation file retains the laboratory detail needed to repeat the accepted procedure. This report uses generic solvent and reagent descriptions without assigning unsupported reference-material results.

9 Sequence construction

Sequences distinguish blank response, reference response, sample response and subsequent checks of system suitability. Suitability was assessed before release and considered across the sequence. A suitable opening check alone cannot justify data acquired after a later failure. Injection order was retained as acquired so that the check-to-sample relationship could be reviewed.

The sequence sheet links acquisition files, preparation entries and calculations. Re-injection retains its reason; an unexplained repeat cannot be selected solely because its value looks preferable. If a sequence is replaced, the earlier acquisition remains with its disposition. The reported table contains released evidence, without treating an ineligible sequence as a second acceptable measurement.

10 System suitability and data eligibility

System suitability addresses separation, repeat response and the ability to measure the required endpoints. Acceptance uses the criteria attached to the controlled method. Samples become eligible for reporting when their applicable checks support interpretation. Suitability assesses the measurement sequence and is not an alternative product specification.

Saira Mahmood checked the suitability decision for the reference and sample data used in the calculations. Chromatographic appearance was considered with the numerical checks, particularly where a shoulder or neighbouring peak could affect integration. The report-level evidence retains the parameter and criterion together; it does not infer a new numerical suitability result from a general approval statement.

11 Calculation and reporting conventions

Assay is reported as percent label claim and related substances as area percent under the qualified conventions. Relative retention assists peak tracking but does not establish chemical identity alone. The original reporting precision and any censoring symbol are preserved. A value below a reporting boundary is not replaced by zero.

Calculations retain dilution, reference correction and source response in the analytical file. Review used this trail before releasing the report table. Units are carried in the headings rather than inferred from neighbouring endpoints. Water, where supplied, uses percent w/w and is not subtracted from assay to create an unqualified corrected value. These distinctions remain part of any transferred evidence.

12 Chromatogram and integration review

Complete traces were reviewed before numerical transcription. Boundaries were applied consistently between reference, challenged or aged material and routine sample preparations. The blank was considered in the same retention region as each reported impurity. An intervention retained the source trace and the reason for the adjusted boundary.

An unexpected peak was not removed merely because it was absent from the reference. Its treatment followed the method reporting conventions. A principal-peak shoulder required a separation and integration assessment instead of an automatic increase in principal-component area. The report assigns no chemical structure from retention alone and does not invent a mechanism for a changing secondary peak.

13 Formulation transfer

The transfer team required a linked account of lot identity, storage history and the suitability of the assay and related-substances procedure. This report supplies that account for CVL tablet lot 3. A transfer decision must retain the tested formulation and container configuration. A new formulation does not inherit the conclusion merely because its active ingredient is unchanged.

The accepted evidence concerns the tested stability lot and the completed interval. A transfer package should cite the approved tables with this conclusion. Extracting an isolated assay value would remove the associated impurity assessment and storage history, both of which are material to the decision.

14 HPLC method qualification — approved evidence

Assay and related-substances validation forms part of WA/2020/313. The table preserves each parameter, result and criterion together. Saira Mahmood reviewed that comparison before scientific release. The numerical summary is read with preparation and acquisition entries; a criterion cannot be detached from the parameter to which it applies.

Validation concerns the tested matrix and controlled procedure. It does not establish suitability for an altered extraction or separation. Any receiving laboratory retains the accepted preparation and reporting conventions or assesses its proposed change. The report does not add an uncommissioned validation parameter or create a new acceptance value.

Table 2. Assay and related-substances validation summary

Parameter	Result	Acceptance
Linearity (r^2)	0.9994	≥ 0.999
Range (% nominal)	50-150	80-120 minimum
Accuracy 80% (% recovery)	100.0	98.0-102.0
Accuracy 100% (% recovery)	100.2	98.0-102.0
Accuracy 120% (% recovery)	100.7	98.0-102.0
Repeatability (%RSD, n=6)	0.46	≤ 2.0
Intermediate precision (%RSD)	0.86	≤ 2.0
LOQ (% of nominal)	0.02	≤ 0.05

15 ICH chamber programme — design

The programme followed CVL tablet lot 3 through a reported interval of 12 months. Long-term and elevated-temperature observations retain their assignments in the approved table. Pulls were assigned before preparation with disposition captured on the batch card. The final timepoint bounds the reported evidence.

Each condition is interpreted within itself before comparisons are made. A short stress exposure is not inserted as a later long-term observation. Storage history remains relevant even when assay response is little changed. A reader must be able to identify condition and timepoint without relying on an unstated default. No missing pull is reconstructed here.

16 Chamber assignment and monitoring

Leon Wu maintained the link between material placed into storage and its assigned condition. Monitoring entries were retained with the protocol. A location change would require continuity of exposure and documentation of any period outside the intended condition. An eligible analytical sequence alone cannot establish that the exposure was maintained.

Review distinguished routine access from an excursion requiring assessment. Impact depends on affected material, timing and endpoint, rather than an alarm entry alone. This report does not reconstruct an unrecorded temperature reading. Chamber labels in the approved table remain the supplied labels, and the report does not create additional numerical storage arms.

17 Stability pulls — removal and reconciliation

Pull instructions were reconciled to the batch card and the intended endpoint before removal. Material remained identifiable through appearance examination and preparation. Remaining units retained their assigned disposition. A pulled container was not described as untouched reserve simply because only a portion had been used.

The sequence sheet identifies the analytical use of removed material. Review checked the intended pull instead of relying on a calendar month in a filename. Pull chronology was considered with interim and completion release gates. A delay affecting exposure or preparation would require assessment before reporting. The source entries remain the basis for that assessment.

18 Condition-specific numerical evidence

The table below reproduces the released condition-specific evidence. Timepoint is in months; assay and impurities retain their original units. The Result column belongs to the analytical table and is not rewritten to force agreement with a broader product decision. Initial entries remain attached to their individual conditions rather than collapsed into a universal baseline.

The accepted evidence concerns the tested stability lot and the completed interval. A transfer package should cite the approved tables with this conclusion. Extracting an isolated assay value would remove the associated impurity assessment and storage history, both of which are material to the decision.

Table 3. Stability pulls with original condition and timepoint labels

Condition	Timepoint (months)	Assay (% LC)	Imp A RRT 0.86 (%)	Imp B RRT 1.12 (%)	Total impurities (%)	Water KF (% w/w)	Result
25 °C/60% RH	0	100.6	0.04	<0.05	0.19	1.8	Complies
25 °C/60% RH	1	100.5	0.05	<0.05	0.20	1.6	Complies
25 °C/60% RH	3	100.1	0.08	<0.05	0.18	0.9	Complies
25 °C/60% RH	6	99.9	0.10	0.06	0.22	2.0	Complies
25 °C/60% RH	9	99.5	0.12	0.07	0.35	1.3	Complies
25 °C/60% RH	12	99.8	0.16	0.08	0.33	2.0	Complies
30 °C/65% RH	0	100.6	0.04	<0.05	0.19	1.8	Complies
30 °C/65% RH	1	100.3	0.06	<0.05	0.21	1.4	Complies
30 °C/65% RH	3	100.6	0.08	0.05	0.23	2.5	Complies
30 °C/65% RH	6	100.3	0.13	0.10	0.32	2.2	Complies
30 °C/65% RH	9	100.3	0.19	0.10	0.36	0.9	Complies
30 °C/65% RH	12	99.6	0.21	0.11	0.45	1.9	Complies
40 °C/75% RH	0	100.6	0.04	<0.05	0.19	1.8	Complies
40 °C/75% RH	1	99.8	0.10	<0.05	0.20	2.5	Complies
40 °C/75% RH	3	99.6	0.19	0.12	0.39	1.5	Complies
40 °C/75% RH	6	98.4	0.34	0.21	0.69	0.9	Complies

Specification: assay 95.0-105.0% LC; Imp A (RRT 0.86, specified) ≤0.50%; Imp B (RRT 1.12, specified) ≤0.30%; any unspecified impurity ≤0.20%; total impurities ≤2.0%

19 Assay through the interval

Assay measures the principal component under the qualified reporting procedure. Comparison uses the condition-specific progression in the table. No averaged baseline is substituted for initial rows. An apparent change is read with preparation control and sequence suitability, rather than interpreted as a standalone product mechanism.

The agreed assay and impurity limits were met for the reported CVL tablet lot through the completed interval. This conclusion is restricted to the tested lot and configuration; it does not release a later manufactured lot.

20 Related-substances profile

Individual impurity columns retain relative-retention labels from the approved evidence. Total impurities remains the reported aggregate under the method convention. A changing column is not converted into a structural identification. Review checks whether the same region remains evaluable as samples age or experience different storage conditions.

A small principal-component change does not explain every secondary peak. The accepted numerical values retain censoring and reporting precision. No mechanism is assigned from relative retention alone, and no unsupported secondary-peak value is added to the table.

21 Water and physical endpoints

Water KF retains the supplied percent w/w heading. Its measurement basis differs from assay and area-percent impurities. These values are not subtracted to create an unqualified corrected assay. Physical examination remains linked to the inspected container and pull, complementing the chemical measurements.

For tablets, moisture context may assist interpretation of storage and packaging but does not independently establish a degradation pathway. The report introduces no numerical appearance score. Physical observations and analytical responses retain their distinct roles in the product assessment.

22 Long-term interpretation

The accepted evidence concerns the tested stability lot and the completed interval. A transfer package should cite the approved tables with this conclusion. Extracting an isolated assay value would remove the associated impurity assessment and storage history, both of which are material to the decision.

The report does not create a later observation by extending a fitted line. Final sample identity and analytical suitability are both required for the endpoint decision. The client may combine this evidence with its product knowledge; additional claims remain outside the observations supplied here. A copied table must preserve the completed interval rather than presenting the endpoint as indefinite evidence.

23 Ambient and elevated-temperature interpretation

Elevated-temperature observations are supporting evidence under their own duration and condition. A stress observation does not establish a rate at a different temperature. A shorter stress interval cannot fill a missing later long-term pull.

Cross-condition comparison retains identity and endpoint definition. Packaging and handling differences are considered before attributing all change to temperature. The scientific conclusion remains at the level supported by the protocol. No kinetic parameter, untested exposure duration or additional quantitative condition is inferred from the qualitative client outcome.

24 Configuration-specific decision

The transfer team required a linked account of lot identity, storage history and the suitability of the assay and related-substances procedure. This report supplies that account for CVL tablet lot 3. A transfer decision must retain the tested formulation and container configuration. A new formulation does not inherit the conclusion merely because its active ingredient is unchanged.

The tested configuration bounds the report. Container, closure and product presentation are not changed by editing a downstream label. Implementing the client decision retains the relevant configuration description. Later changes require assessment of the changed material against the historical evidence. The report does not create a new packaging or handling specification merely through inclusion in a transfer file.

25 Suitability at the final pull

Final-pull results used analytically eligible evidence. Suitability was considered with earlier checks. An initial method conclusion would not excuse an unresolved interference in aged material at completion.

Review considered whether evolving secondary peaks compromised assay or impurity measurement before drawing the product conclusion. A reported final result does not itself create a numerical method-performance value. The accepted method table supplies that evidence where commissioned. For the chamber-only scope, sequence checks support measurement eligibility without implying a separate validation package.

26 Endpoint reconciliation at completion

The completion release gate on 30 Nov 2021 brought together storage history, pull identity and released analytical evidence. The final reported interval is 12 months. Report release followed that evidence close, rather than creating a further exposure interval.

Reconciliation considered the condition-specific table and the product observation used for the client decision. The initial analytical rows are not silently replaced by a project-level summary baseline. Analytical Result labels are retained, even where a broader appearance or packaging conclusion requires separate interpretation. This preserves what was measured and what was decided without conflating the two.

27 Appendix A — pull trace for reuse

A reused row retains accession, product, condition, timepoint, preparation and eligible sequence. A summary can shorten narrative but must not reduce the trace to a product name followed by an unexplained number. Follow the reported endpoint back to removal and acquisition, then check the forward path from receipt and placement.

An incorrect condition assignment can survive a perfectly sound chromatographic calculation. The trace therefore carries the exposure basis as well as the response calculation. The approved table remains the numerical source; this appendix supplies its use context without adding a later pull.

28 Appendix B — unit and condition key

Temperature uses °C; relative humidity remains in the condition label where supplied. Timepoint uses months from placement. Assay uses percent label claim, impurities area percent and water percent w/w. These conventions remain attached to exported data.

A missing condition-specific observation is not filled from a neighbouring row. Reporting-boundary symbols are preserved when formats change. A trend display should retain condition grouping and endpoint units. No table row is relabelled to create an arm absent from the supplied evidence.

29 Appendix C — final-pull inquiry

A final-pull inquiry retains removal history, preparation and suitability together. The completion date alone does not demonstrate exposure duration. If a copy loses its storage label, it no longer establishes the intended condition-specific conclusion.

Begin with the final row, then reconcile condition and timepoint to the batch card. Return through preparation and acquisition to the released table. This appendix describes use of the completed evidence; it does not assert a newly measured observation after the completion release gate.

30 Appendix D — interpretation safeguards

The accepted evidence concerns the tested stability lot and the completed interval. A transfer package should cite the approved tables with this conclusion. Extracting an isolated assay value would remove the associated impurity assessment and storage history, both of which are material to the decision.

Method performance supports measurement; it does not replace an unacceptable product observation. An expected product trend does not justify accepting an unsuitable sequence. These distinctions remain visible in the client summary so that a downstream user can tell the numerical evidence from the product-level decision.

31 Appendix E — client package contents

The package for Corvenlea Pharmaceuticals plc contains this approved report, the released numerical evidence and the controlled procedure context. Project identity is the package reference; the accession remains the laboratory source-data reference. Neither is replaced by an informal product label.

The package supports the recorded decision. A later user seeking a different claim needs evidence appropriate to that claim. The approved proposal for WA/2020/313 remains the commercial scope reference; it is not experimental evidence.

32 Appendix F — reading order

Read protocol and product identity first, then condition and timepoint, then endpoint and unit. Consider sequence eligibility and any material physical observation before reading the scientific conclusion. This order prevents a favourable isolated value from being detached from its interpretation basis.

The transfer team required a linked account of lot identity, storage history and the suitability of the assay and related-substances procedure. This report supplies that account for CVL tablet lot 3. A transfer decision must retain the tested formulation and container configuration. A new formulation does not inherit the conclusion merely because its active ingredient is unchanged.

33 Deviations and disposition

Final release contains no unresolved deviation changing the stated conclusion. This report does not introduce an extra assay value to stand for a correction or repeat. A deviation note belongs with its source acquisition or preparation and the associated impact decision. Ineligible evidence is not made acceptable by losing its history in a copied table.

Method development is distinguished from a failure of the final controlled procedure. A selectivity change belongs in the method history; an event affecting an accepted sequence requires its own impact assessment. The final conclusion uses released evidence within the completed scope.

34 Scientific conclusion and client disposition

The tablet lot remained within the agreed assay and impurity limits, and Corvenlea Pharmaceuticals plc used the report in its formulation transfer package.

The accepted evidence concerns the tested stability lot and the completed interval. A transfer package should cite the approved tables with this conclusion. Extracting an isolated assay value would remove the associated impurity assessment and storage history, both of which are material to the decision.

Alice Penrose considers the evidence sufficient for the stated decision at completion on 30 Nov 2021. No observation beyond the tested interval or outside the method scope is asserted. Future material and future sequences require their own identity and suitability checks. The protocol outcome is retained with the underlying evidence under WST-20-009708.

35 Archive index and approval basis

The report is filed under WA/2020/313 and WST-20-009708. The associated result file is CVL_PULLS_AND_SUITABILITY_RESULTS_Rev03.xlsx, supplying the sequence and endpoint trail behind this report.

The approval package retains receipt entries, preparation trails, source chromatograms, calculations and suitability decisions. Leon Wu checked sample linkage; Saira Mahmood checked analytical evidence; Alice Penrose approved the conclusion. An extracted table should carry project, product, units, condition where applicable and approved issue status. The retained copy bears the original issue date.

Approved archive navigation

Component	Purpose	Link
Batch card	Identity and handling	Accession and product
Sequence sheets	Acquisition and suitability	Preparation and injection order
Calculations	Result derivation	Response and dilution trail
Report	Decision and limits	Project and issue date

Name	Role	Signature	Date
Leon Wu	Stability Coordinator — sample traceability	<u>L Wu</u>	02 Dec 2021
Saira Mahmood	QC Method Validation Lead — analytical review	<u>S Mahmood</u>	02 Dec 2021
Mark Campbell	QA Officer	<u>M Campbell</u>	02 Dec 2021
Alice Penrose	Head of Analytical Services — approval	<u>A Penrose</u>	02 Dec 2021